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Effects of massed and spaced task repetitions on L2 writing task performance and task emotions

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Abstract

Recently, in task repetition research, distributed practice has received considerable attention for its potential impact on learners' performance (Rogers, 2022). While several studies have begun exploring its effects on oral task repetition, little attention has been paid to the effects of spacing in writing task repetition and its interaction with learners' emotional states (i.e., enjoyment and anxiety). Focusing on 270 learners studying English as their second language (ESL) in the U.S., the current study examined their writing performance and emotions in the same decision-making task under three different schedules: (1) in one class (massed practice), (2) with a two-day interval (short-spaced practice), and (3) with a one-week interval (long-spaced practice). Following each writing session, learners completed a questionnaire assessing their anxiety and enjoyment during the task. The results showed that the long-spaced group outperformed the other groups after the experiment in general syntactic complexity, phrasal complexity, and lexical diversity, and they also experienced lower anxiety and higher enjoyment. Conversely, the massed group had lower achievements in phrasal complexity and lexical diversity and experienced lower enjoyment and higher anxiety than the other groups. These findings suggest that the interval between task repetitions can significantly influence learners' writing performance and emotional states.

Keywords: Massed and spaced task repetitions · Linguistic performance · Task-specific emotions · TBLT (Task-Based Language Teaching) · Individual differences · L2 writing

Introduction

In second language (L2) classrooms, task repetition plays a crucial role in enhancing both learners' task performance and L2 development. Recognizing its significance, a substantial body of empirical research has investigated task repetition's impact on language performance from various angles. These studies explore the effects of exact task repetition and procedural repetition (e.g., Kim & Tracy-Ventura, 2013), additional interventions accompanying task repetition (e.g., Tran & Saito, 2021), and the influence of time pressure (e.g., Thai & Boers, 2016). Recent inquiries within this research domain have focused on the distribution of spacing —a well-studied topic in cognitive psychology—highlighting its potential to enrich our understanding of task repetition (Bygate, 2018; Rogers, 2022).

The distribution of spacing refers to the timing between repetitions (Rogers, 2017) and typically encompasses two phenomena: spacing effects (massed vs. spaced practice) and lag effects (short-lag vs. long-lag; e.g., Cepeda et al., 2006). Spacing effects underscore the advantages of spaced practice over massed practice. Massed practice involves repeated tasks without temporal intervals, whereas distributed practice spaces out repetitions, allowing time between performances—such as completing a task and revisiting it a day or a week later. Lag effects examine how short versus long intervals between practices affect learning outcomes. In line with cognitive psychological studies, several inquiries have found the effects of spacing distribution on L2 oral performance and development (Suzuki & Hanzawa, 2022; Bui et al., 2019; Kakitani & Kormos, 2024; Kobayashi, 2021; Suzuki, 2021). This study extends this line of research by investigating the effects of distributed practice on writing tasks. While task repetition's effects on oral performance have received considerable attention, research on writing performance remains limited (e.g., Ajabshir & Poorebrahim, 2021; Sánchez et al., 2020). Given the distinct characteristics between writing and speaking such as time on task and the opportunity for revision (Kormos, 2014; Manchón, 2014; Tavakoli, 2014), exploring the impact of distributed practice on L2 written production is essential.

Furthermore, in Task-based Language Teaching (TBLT)—an instructional approach that promotes language learning by engaging students in practical, real-world tasks—research has focused on understanding how different task features (Abdi Tabari 2020; Révész et al., 2017) and individual cognitive differences among learners (Skehan, 2024) affect task performance. Recently scholars in this domain have begun paying much attention to the emotional dimensions of L2 writing task performance in both positive and negative ways (Abdi Tabari et al., 2024; Cheong et al., 2023). Learners' emotional states during a task have been found to play a crucial role in their L2 learning (Hiver et al., 2024; Pilotti et al., 2023). While positive emotions during the task can enhance their active involvement which leads to successful

learning, negative emotions can cause disengagement from the task (Skinner et al., 2008). Despite numerous studies examining linguistic outcomes of task repetition (Hidalgo & Garcia Mayo, 2021; Hu, 2018), to our knowledge, very limited research has explored emotional effects related to task repetition schedules. Therefore, integrating emotional responses as a non-linguistic goal within TBLT is crucial.

Building on these gaps, the current study aims to compare the effects of three different repetition schedules on L2 writing task performance, measured in terms of syntactic complexity, accuracy, lexical complexity, and fluency, collectively referred to as CALF. CALF encompasses four key dimensions: *syntactic complexity*, which reflects the size, elaborateness, richness, and diversity of the learner's linguistic L2 system; *accuracy*, defined as the degree of deviancy from a particular norm, where deviations are typically characterized as errors; *lexical complexity*, which evaluates the diversity and sophistication of vocabulary employed by writers; and *fluency*, characterized by the ease, eloquence, and smoothness of writing, particularly in terms of the number of words produced within a given time frame (Abdi Tabari, 2021, 2022). By examining these four metrics, the study seeks to provide insights into how different repetition schedules influence L2 learners' writing task performance. In addition, it will explore two task-specific emotions: anxiety and enjoyment. This research aims to investigate how different task repetition schedules impact both L2 writing task performance and task-specific emotions across specific time periods.

Review of literature

Distributed practice: cognitive psychology insights and L2 research findings

Cognitive and educational psychology have been developing and evaluating effective learning strategies to optimize learners' learning experiences. One of those strategies is distributed practice effects (e.g., Cepeda et al., 2006), which encompasses both spacing effects and lag effects. Spacing effects are well-documented (Donovan & Radosevich, 1999) across various subjects and skills such as mathematics, typing, and verbal memory. On the other hand, findings of lag effects are mixed and challenging to define and demonstrate (Cepeda et al., 2008). Several theories have been proposed to explain these distributed effects (see Kim & Webb, 2022 for a review). For example, the desirable difficulty framework states that spacing makes learning more challenging, thereby encouraging learners to exert maximal effort to remember what they have learned, which leads to improved learning and memory retention (Bjork, 1994; Suzuki et al., 2019). Meanwhile, the deficient processing hypothesis suggests that spaced practice can lead to more efficient processing and better memory retention than massed practice (Toppino & Bloom, 2002).

Recently, researchers have begun investigating the effects of distributed practice on L2 learning, particularly in vocabulary and grammar. For example, Nakata and Elgort (2020) found that learners engaged in spaced practice outperformed those in massed practice in terms of meaning recall and the meaning-form matching posttests. Similarly, Miles (2014) demonstrated that spaced practice led to better acquisition of grammatical structures than massed practice. However, results concerning the lag effect were mixed, mirroring findings in cognitive psychology. For instance, while Rogers (2015) reported that longer-spaced practice was more effective than short-spaced practice for grammar learning, Suzuki (2017) observed the opposite effect, and Kasprowicz et al. (2019) found no significant difference between longer and shorter spaced practice. To address these inconsistencies, Kim and Webb (2022) conducted a meta-analysis of 48 studies in L2 research, revealing that spaced practice significantly enhances L2 learning and retention compared to massed practice, with a medium-to-large effect size. Their findings also indicated that while shorter spacing was as effective as longer spacing in immediate posttests, longer spacing proved more beneficial in delayed posttests. Notably, they identified that the type of activity—such as paired association, comprehension, and production tasks—moderated the effects of distributed practice. Specifically, for production tasks, shorter spacing was more advantageous in immediate posttests.

Research on task repetition and its impact on L2 writing task performance

Task repetition is defined as the "repetition of a given configuration of purpose, and a set of content information" (Bygate, 2018, p. 2). Research in this area has extensively focused on oral performance and demonstrated the facilitative role of various L2 outcomes (e.g., Ahmadian, 2011; Gass et al., 1999; Lambert et al., 2017). The beneficial effects of task repetition are often attributed to the redirection of learners' attentional resources during task repetition. Based on the speech production model (Levelt, 1989), Bygate (2001) suggests that during the initial performance, learners primarily focus on conceptualization (generating the content of the message). However, as they become more familiar with the content through repeated task engagement, they increasingly focus on formulation (linguistic encoding) and articulation (producing sounds), potentially leading to more proficient processing and enhanced L2 development. Many researchers in this field aim to explore how task repetition affects learners' ability to perform specific tasks and how these improvements relate to broader L2 development (Bygate, 2018).

Researchers have recently begun incorporating distributed practice effects into task repetition studies, significantly enriching this field of inquiry (Suzuki & Hanzawa, 2022; Bui et al., 2019; Kakitani & Kormos, 2024; Kobayashi, 2021; Suzuki, 2021). More recently, Rogers

(2022) summarized findings in cognitive psychology, emphasizing the impact of distributed practice effects on task repetition research. Among them, two studies particularly relevant to the current investigation are those by Kobayashi (2021) and Suzuki and Hanzawa (2022). In his study, Kobayashi (2021) assigned Japanese learners of English to two different practice schedules (massed and spaced practices) for an oral narrative task. The results indicated varying effects of spacing on learners' oral performance across four indices—complexity, accuracy, fluency, and lexical variety. Notably, while there were no significant differences in the first and second performances in the four indices, learners under spaced practice showed significant improvement in lexical variety in a new task administered one week after the task repetition. Similarly, Suzuki and Hanzawa (2022) explored both spacing effects (massed vs spaced) and lag effects (short-spaced and long-spaced). They divided learners into three groups: massed practice, short-spaced practice, and long-spaced practice, assessing performance through six iterations of oral narratives. Their findings revealed that while massed practice enhanced fluency in the immediate posttest, no significant differences were observed across the schedules in the delayed posttest.

Taken together, the findings from previous oral task repetition studies suggest that the interval between repetition influences learners' task performances with various outcome variables. These results, in turn, suggest that spacing should be considered a critical variable in task repetition research (Rogers, 2022).

Optimizing L2 writing task performance through systematic task repetition schedules

The current study expands research on the effects of distributed practice on writing tasks. While task repetition has received considerable attention, most studies have primarily focused on the oral domain, with fewer exploring its impact on writing (e.g., Abdi Tabari et al., 2024; Hidalgo & García Mayo, 2021; Y. Kim et al., 2022; Nitta & Baba, 2014). For example, Amiryousefi (2016) examined the effects of different task repetition types (exact repetition vs. procedural writing repetition) on repeated picture narration and noted improvements in accuracy and fluency in the exact repetition group. Similarly, Fukunaga's (2023) study involved learners engaging in descriptive and argumentative writing tasks five times, with significant enhancements in fluency and syntactic complexity observed in the descriptive task. These findings underscore the potential benefits of task repetition for enhancing wiring performances.

Oral and writing productions share many similarities in language processing, and several studies on writing task repetition have used Bygate's (2001) rationale as their theoretical foundation (e.g., Amiryousefi, 2016; Hidalgo & Lázaro-Ibarrola, 2020). However, significant differences exist between oral and written productions (Kormos, 2014; Tavakoli, 2014). One

of those differences is time-on-task. In writing, learners have more time to engage in a task, allowing them to distribute their attentional resources across various task demands simultaneously (Manchón, 2014). Another distinction is the opportunity for revision during a task; unlike in oral production, writers can review and revise their output as they work as they engage in the task (Nitta & Baba, 2014). Sánchez et al. (2020) highlighted the differing effects of modality on task repetition performance, showing that while fluency improved in oral repetition tasks, no significant changes were observed in written repetition tasks. However, notably, learners in writing task repetition produced more complex sentences than those in oral task repetition. Although previous research on writing task repetition has enhanced our understanding of task repetition dynamics, to the best of our knowledge, no study has systematically manipulated repetition intervals (i.e., investigated the distributed effect) in writing. Given the differences between oral and writing productions, it is unlikely that findings from oral task repetition studies can be directly applied to writing. Accordingly, the current study aims to address this significant gap by examining the effects of distributed practice effects (spacing effects and lag effects) on writing task performance.

Anxiety and L2 writing task performance

In the field of second language acquisition (SLA), anxiety has long been a focal point among researchers exploring individual differences (MacIntyre & McGillivray, 2023). Among these, L2 writing anxiety has garnered particular attention due to its significant impact on language performance. Defined by Cheng (2004), L2 writing anxiety encompasses persistent feelings of apprehension, heightened physiological arousal and avoidance behaviors specifically related to writing tasks in a second language. This conceptualization draws from Lang's (1971) model of anxiety, which categorizes it into cognitive anxiety (worry and fear of evaluation), somatic anxiety (physical arousal), and avoidance behaviors.

The Second Language Writing Anxiety Inventory (SLWAI), developed by Cheng (2004), has spurred investigations into how L2 writing anxiety influences achievement. Empirical evidence consistently shows a moderate negative correlation between L2 writing anxiety and proficiency (Abdi Tabari et al., 2025; Rahimi & Zhang, 2018; Zhang & Zhang, 2022), indicating that higher levels of anxiety are associated with poorer writing performance across various measures. However, the exact direction of this relationship remains underexplored, necessitating further research into its nuanced impacts on language production. Furthermore, while studies examining the effects of L2 writing anxiety on task performance are relatively scarce (Güvendir & Uzun, 2023; Tahmouresi & Papi, 2021; Zabihi et al., 2020), there is a critical need to understand how task characteristics, particularly task repetition schedules, may influence anxiety levels and, consequently, performance. A recent meta-analysis on L2 anxiety

and performance underscored the lack of focus on L2 writing anxiety due to limited available research (Zhang, 2019). This gap highlights the need for more comprehensive investigations that delve into how anxiety influences various language tasks and the mechanisms through which it operates.

Several studies have provided insights into the effects of L2 writing anxiety on performance outcomes. For instance, Saeedi and Farnia (2017) found significant negative correlations between L2 writing anxiety and measures of complexity, accuracy, and fluency (CAF) in timed narrative tasks. These findings suggest that heightened anxiety levels can adversely affect linguistic fluency and accuracy in writing. Rahimi and Zhang (2018) explored the impact of task complexity, revealing negative associations between avoidance behaviors and syntactic complexity in argumentative essays. This indicates that anxiety-related avoidance tendencies could hinder the development of syntactic complexity, particularly in more demanding writing tasks. Zabihi et al. (2020) reported differing impacts of anxiety across task genres, with cognitive anxiety particularly affecting performance metrics in both narrative and argumentative tasks.

Despite these findings, research specifically examining how L2 writing anxiety interacts with task repetition schedules remains very limited. This gap in the literature impedes a clear understanding of how anxiety influences pedagogical interventions, such as distributed practice, which are essential for optimizing L2 writing performance over time. From a cognitive psychology perspective, task repetition can reduce cognitive load and enhance familiarity with writing tasks, thereby potentially alleviating anxiety. As learners repeatedly engage with similar tasks, they may experience increased self-efficacy and reduced fear of evaluation, which can translate into improved writing performance. Therefore, it is necessary to prioritize the integration of L2 writing anxiety into investigations of task repetition schedule effects. This approach will shed light on its potential contributions to both task performance outcomes and emotional responses in L2 writing contexts.

Enjoyment and L2 writing task performance

In recent years, SLA research has shifted to include both negative and positive emotions, expanding beyond the traditional focus on anxiety and other negative emotions (Dewaele & MacIntyre, 2014). Enjoyment, defined as a complex emotion involving pleasure and satisfaction during language learning activities, has emerged as a significant factor influencing motivation and learning outcomes (Jiang & Dewaele, 2019). According to Teimouri (2017), L2 enjoyment encompasses the positive emotions learners experience while engaging with the target language, whether in structured classroom environments or authentic real-life contexts.

The interplay between task repetition and emotional responses, particularly enjoyment, is crucial to understanding how learners engage with L2 writing. Seminal research by Dewaele and MacIntyre (2014) revealed a negative correlation between language enjoyment and anxiety among learners, indicating that these emotional states are distinct and inversely related. Their findings suggest that students typically experience higher levels of enjoyment than anxiety in language classes, with factors such as multilingualism, age, and educational level positively influencing enjoyment levels. As learners encounter repeated writing tasks, the potential for increased familiarity and reduced cognitive load may enhance their enjoyment while simultaneously lowering anxiety levels. This effect is particularly relevant in the context of task repetition schedules, which can promote a sense of mastery and reduce apprehension in language performance.

Supporting these findings, Dewaele and Alfawzan (2018) examined foreign language enjoyment and anxiety among L2 learners in London secondary schools and EFL learners in Saudi Arabia. Their study confirmed that higher levels of language enjoyment positively impact language performance, while anxiety negatively affects it. Similarly, Khajavy et al. (2018) found that Iranian English learners generally experience more enjoyment than anxiety in L2 classes, with enjoyment significantly predicting a greater willingness to communicate in English. This underscores the importance of structuring task repetition to maximize enjoyment, as positive emotional experiences can foster greater engagement and willingness to communicate, further enhancing L2 performance.

The dynamic nature of language enjoyment has been explored in longitudinal studies. Dewaele and Dewaele (2017) observed that L2 learners' enjoyment fluctuates due to socio-psychological and environmental factors, highlighting the variability of emotional experiences over time. Elahi Shirvan and Talebzadeh (2018) further examined these dynamics, contributing to our understanding of how contextual factors shape emotional responses in language learning environments. These fluctuations suggest that task repetition schedules, which can vary in their design and implementation, may play a significant role in shaping L2 learners' emotional experiences. By systematically varying the nature and frequency of task repetitions, educators can potentially influence the emotional climate of the learning environment, thereby impacting enjoyment and anxiety.

Tahmouresi and Papi (2021) identified positive associations between the ideal L2 writing self and L2 writing enjoyment and achievement, although L2 writing enjoyment was not directly related to L2 writing achievement. In exploring the connection between enjoyment and the ideal L2 self, Papi and Khajavy (2021) found that the ideal L2 self positively correlates with enjoyment. Additionally, Dewaele and Alfawzan (2018) noted that positive emotional states positively correlate with L2 learners' language performance. Enjoyment in L2 writing tasks

enhances curiosity about the L2 and its cultural contexts, builds resilience, leverages multiple intelligences, and fosters an empathetic and determined approach to learning (Byrd & Abrams, 2022). Learners who enjoy writing in the L2 are more likely to set challenging goals, monitor their progress, and persist despite difficulties (Hyland, 2003). Recent research by Abdi Tabari et al., (2024) and Zhu et al. (2022) provides further evidence on the relationship between enjoyment and L2 writing task performance. These studies consistently show positive correlations between enjoyment and improved writing outcomes. However, the challenge remains in understanding the precise mechanisms through which enjoyment influences L2 writing performance. Exploring how task repetition schedules interact with enjoyment and anxiety is essential to optimize L2 writing outcomes in the TBLT context.

The current study

Despite extensive research on task repetition's effects on oral performance in second language acquisition, a significant gap remains in understanding its impact on writing performance and emotional states. Existing studies have primarily focused on exact and procedural repetition, additional interventions, and the influence of time pressure, often neglecting the distribution of spacing—a crucial element in cognitive psychology that could enhance our comprehension of task repetition (Bygate, 2018; Rogers, 2022). Furthermore, while the cognitive aspects of task performance have been well-explored, the emotional dimensions, particularly in the context of L2 writing tasks, have been largely overlooked. Positive and negative emotions, such as anxiety and enjoyment, can profoundly influence L2 learners' writing performance, yet these factors have not been sufficiently integrated into TBLT research. To address these gaps, the current study aims to compare the effects of three different repetition schedules on L2 writing task performance as well as the task-specific emotions of anxiety and enjoyment. Additionally, this research seeks to examine how these repetition schedules influence both performance and emotional responses over specific time periods, thereby providing a more comprehensive understanding of task repetition in a task-based L2 writing context. To achieve these purposes, the following research questions and hypotheses were formulated:

RQ 1: How do different schedules of task repetition affect L2 writing task performance and emotion?

Hypothesis #1. We hypothesize that long-spaced, short-spaced, and massed repetitions differentially affect L2 writing task performance and emotion.

Hypothesis #2. For task performance, we hypothesize that learners in the long- and short-spaced repetition groups will outperform their peers in the massed repetition group in terms of CALF. Furthermore, we hypothesize that learners in the long-spaced repetition group will outperform their peers in the short-spaced repetition group in terms of CALF.

Hypothesis #3. For task emotion, we hypothesize that learners in the long- and short-spaced repetition groups will have lower anxiety and higher enjoyment than their counterparts in the massed repetition group. Furthermore, we hypothesize that learners in the long-spaced repetition group will have lower anxiety and higher enjoyment than their peers in the short-spaced repetition group.

RQ 2: How do different schedules of task repetition affect changes in L2 writing task performance and emotion over time?

Hypotheses #1. We hypothesize that different schedules of task repetition would differentially affect changes in L2 writing task performance and emotion over time.

Hypotheses #2. We hypothesize that overall all learners will improve their task performance, experience lower anxiety, and have higher enjoyment as they repeat their tasks. Therefore, their task performance and emotional state will be most desirable after two repetitions.

Hypotheses #3. In terms of the differential impacts of task repetition schedules on change, we hypothesize that the effects of each task repetition schedule will vary across different time periods, such that the change in effects of long-spaced task repetition will be the largest, followed by short-spaced and massed repetition schedules.

Methods

Study design

To answer the research questions, we designed a true experiment with three groups, each corresponding to a different task repetition schedule: long-spaced, short-spaced, and massed repetition. For the long-spaced repetition schedule, the group completes the L2 writing task in the first week and then repeats it once in each of the following two weeks (e.g., Weeks 6, 7, and 8). For short-spaced repetition, the group performs the task on the first day and repeats it twice every two days, completing the repetition in five days (e.g., Monday to Friday). For massed repetition, the group performs the initial task and repeats it twice immediately after, completing the repetition within a single class meeting. The study design is summarized in Table 1.

Table 1
Study design

	Monday	Wednesday	Friday
Week 6	Massed IT1-IT2-IT3		
	Short-spaced IT1	Short-spaced IT2	Short-spaced IT3
	Long-spaced IT1		
Week 7	Long-spaced IT2		
Week 8	Long-spaced IT3		

IT stands for iteration, so IT1, IT2, and IT3 are for the first iteration, second interaction, and third iteration of the task

Participants and the learning context

This study involved 270 ESL students enrolled in various sections of a first-year undergraduate ESL writing course at a major university in the United States, selected through convenience sampling. The participants comprised 120 males and 150 females, aged between 18 and 22 years, with a mean age of 20.5 years. They represented a diverse array of first language (L1) backgrounds: Arabic (15%), Dutch (10%), Japanese (10%), Chinese (15%), French (10%), Spanish (15%), Korean (10%), Indian (10%), and Portuguese (5%). The majority of students (85%) had been residing in the United States for less than 18 months, with the remaining 15% having lived in the country for three to five years.

Each participant's University Writing Placement Score ranged from 7 to 8 (out of 9), indicating their initial writing proficiency upon entering the university writing program. Before enrolling in the current course, participants had completed an average of 2.5 ESL writing courses, with some having no previous courses and others having completed up to five. Participants had approximately 30 h of classroom instruction at the university before data collection commenced. To ensure comparability in general proficiency levels, participants were required to take the IELTS Task Version 2 writing test at the start of the study, achieving scores between 6.5 and 7.5 (out of 9), which equates to the Advanced High level on the ACTFL scale.

Materials

Writing task

This study utilized a persuasive writing task to examine the writing performance of first-year ESL students. This task was selected because it encourages students to engage with complex issues, promoting higher-order thinking skills as they evaluate different perspectives and construct well-reasoned arguments. Persuasive writing also allows students to practice essential writing skills such as structuring essays, employing persuasive language, and refining their ability to persuade an audience effectively.

Participants were presented with the following scenario:

Your university is considering increasing funding and support for study abroad programs. As a student representative, you are asked to write a persuasive essay addressed to the university administration. In your essay, discuss the potential benefits and challenges of studying abroad, drawing on hypothetical examples and relevant evidence to support your argument. Conclude with a recommendation on whether the university should proceed with increasing funding and support for these programs.

This prompt was specifically designed to be relevant to students' academic context, encouraging engagement with a topic of both personal and institutional significance. The task required students to articulate their thoughts clearly, organize arguments logically, and employ persuasive techniques, aligning closely with essential components of L2 writing task performance. Although the study utilized linguistic metrics such as complexity, accuracy, lexis, and fluency (CALF) as indicators of language performance, these measures do not directly evaluate global text quality or persuasive coherence. Instead, they serve as proxies to assess aspects of linguistic development that may contribute to overall text quality. The focus on these areas allowed us to examine how task repetition influences core language competencies at a granular level, providing insights into the incremental development of L2 writing skills. Additionally, the persuasive writing task aligns with the core objectives of the ESL writing curriculum, promoting skills such as collaboration and peer feedback that are crucial for effective communication in the ESL academic setting.

The task emotion questionnaires

To effectively capture both anxiety and enjoyment in their immediate and situated contexts over time, we utilized Jiang's (2021) task anxiety and task enjoyment scales, which were specifically adapted for the writing task (see Appendix A for the complete list of items). The task enjoyment scale consisted of 5 items, each rated on a 6-point Likert scale ranging from 1 (not at all) to 6 (very much). One example item is "*I enjoyed doing the task.*" Meanwhile, the task anxiety scale contained 6 items, also rated on a 6-point Likert scale from 1 (strongly disagree) to 6 (strongly agree). An example item from this scale is "*When completing the writing task, I felt my heart was pounding.*" The reliability of these scales was confirmed, with Cronbach's alpha coefficients of 0.96 for the task anxiety scale and 0.95 for the task enjoyment scale, demonstrating high internal consistency for both measures. The participants' responses to the various items for each construct were averaged to calculate composite scores for task anxiety and task enjoyment.

Procedure

After obtaining approval from the Institutional Review Board (IRB), we initiated contact with the director of the Core Writing program to outline the study's objectives and procedures comprehensively. Subsequently, we approached instructors of first-year ESL writing classes to secure their cooperation in facilitating the study's logistics. The first author administered both the writing task and post-task emotion questionnaires during scheduled visits to the classes, ensuring that participants were fully informed about the study's purpose, procedures, and their rights to voluntary participation and confidentiality.

To ensure homogeneity in English writing proficiency, participants were administered the IELTS Task Version 2 writing test in Week 5. Following this assessment, participants were randomly assigned to three groups of equal size (n = 90), each assigned a specific task repetition schedule. All writing tasks were typed, consistent with typical course practice in these ESL writing classes. The long-spaced repetition group completed the persuasive-writing task for 35 min on the first day and repeated it twice weekly within the same timeframe over three weeks (Weeks 6 through 8). The duration of 35 min was determined based on pilot study results indicating optimal task completion times for similar participants at the same English proficiency level. Participants in the short-spaced repetition group performed the task within 35 min on the first day and repeated it every two days within the same timeframe, completing their repetitions over five days (Monday to Friday of Week 6). Finally, participants in the massed repetition group completed the initial task within 35 min and immediately repeated it twice within a single class session, without access to dictionaries or other resources. On average, the essays produced were 367 words long (SD = 69), with essay lengths ranging from 309 to 458 words. Following each writing task session, participants completed post-task anxiety and enjoyment questionnaires, each requiring approximately seven minutes to complete. Responses were collected electronically through the Qualtrics platform to ensure efficient data management and analysis.

CALF metrics

To analyze L2 writing samples, we applied the CALF framework, which stands for syntactic Complexity, Accuracy, Lexical complexity, and Fluency. The CALF framework is a conceptual model that categorizes essential dimensions of L2 written production. It plays a significant role in L2 writing research by providing a detailed approach to gauging and understanding the multifaceted nature of L2 writing task performance (Abdi Tabari, 2023). Each dimension was assessed using specific, well-established metrics in this study. Syntactic complexity was evaluated using the mean length of T-unit (MLT) as a general complexity measure, the ratio of dependent clauses to T-units (DC/T) to measure subordination

complexity, and the ratio of complex nominals per T-unit (CN/T) for phrasal complexity. According to Norris and Ortega (2009), syntactic complexity involves assessing general complexity, subordination via dependent clauses, and phrasal complexity via complex nominals within subclauses. MLT was computed by dividing the total number of words by the total number of T-units, where each T-unit includes a main clause plus any attached subordinate or non-clausal structure (Hunt, 1970). Subordination complexity was assessed using DC/T, and phrasal complexity via CN/T (Lu, 2010). Automated computation of these syntactic complexity measures was facilitated by the L2 Syntactic Complexity Analyzer (L2SCA) developed by Lu (2010), ensuring consistency and reliability as validated in previous L2 writing studies. Bivariate correlations among these complexity metrics indicated their non-redundancy ($r < 0.70$).

Accuracy was determined based on error-free clauses, measured as the proportion of error-free clauses in the text. Errors included syntactic, morphological, and lexical categories, excluding non-grammatical errors following Wigglesworth and Storch (2009) as they do not affect meaning. A subset of essays (50%) was coded and scored for accuracy measures by an expert colleague, demonstrating high inter-coder agreement for error-free clauses (EFCs; $r = 0.94$) and correct verb forms (CVFs; $r = 0.96$). For lexical diversity, we utilized the measure of textual lexical diversity (MTLD) recommended by McCarthy and Jarvis (2010). MTLD calculates the mean length of word strings that maintain a specified level of lexical variation, independent of text length, thus capturing unique lexical information. Lexical sophistication was assessed using the log frequency of content words from the CELEX database (WRDFRQmc), selected for its sensitivity to both minor and major gains in writing proficiency (Kyle & Crossley, 2015). Finally, L2 writing fluency was operationalized as the rate of production, measured in words per minute (W/M), following Wolfe-Quintero et al. (1998), and endorsed by Kellogg (1996) as a valid indicator of product-based writing fluency.

Data analysis

We used the Stata version 16 (StataCorp, 2019) as the primary statistical software for data analysis. To begin with the data analysis, we computed descriptive statistics for variables of our interests, including their means, standard deviations, and ranges (i.e., minimum and maximum values). Then, before answering the first research question, we checked the homogeneity between treatment groups after the random assignment by conducting ANOVA with each of the three variables we collected before the experiment, and they were placement test results, age, and gender. To answer the first research question, we ran multiple regression analyses for each of the dependent variables, such as eight CALF metrics and two emotion

variables, at the 2nd and 3rd iterations, which were for measuring the impact of different repetition schedules for one repetition and two repetitions, respectively, along with their corresponding variables at the 1st iteration as well as placement test results, age, and gender variables as control variables. To answer the second research question, we first ran a series of t-tests to check how the experiment influenced our participants' L2 writing task performance and task emotions after one repetition (i.e., comparisons between 1st and 2nd; at the 2nd iteration) and two repetitions (i.e., comparisons between 1st and 3rd; at the 3rd iteration). These changes were investigated for each task repetition schedule. It should be noted here that we understand the importance of adjusting for Type I error rates when conducting multiple t-tests. Although our analysis consists of independent comparisons across 10 different measurements, we applied the Bonferroni correction to ensure the robustness of my findings and maintain the overall significance level (e.g., $p < 0.005$ for statistical significance instead of 0.05; i.e., 0.05/10 = 0.005). This will help mitigate the risk of false positives in our results. Furthermore, we ran a time series cross-sectional (TSCS, see Shadish et al., 2002 for details of this approach) analysis to explore how the effects of task repetition schedules on task performance and task emotions varied over different time periods by including interaction terms between the task repetition schedules and the iterations.

Results

Descriptive statistics

Before delving into data analyses to answer the research questions, we computed descriptive statistics for variables of our interests to understand the collected data. As the data collection was conducted at three different time points, Table 2 represents the descriptive statistics at each iteration. At the 1st iteration (see Table 2), given the emotion was measured by a 6-point Likert Scale, the participants were overall neutral (i.e., between 3 and 4) in terms of their anxiety and enjoyment. In the second and third iterations, we found that the descriptive statistics of variables were largely similar to those in the first iteration. Although the differences were not substantial, in terms of syntactic complexity, accuracy, and lexical complexity, the means of MLT, DCT, CNT, EFC, CVFs, MTLD, and WRDFRQmc were slightly greater at the 2nd iteration than at the 1st iteration, and greatest at the 3rd iteration. However, concerning fluency, the mean of WM became smaller at the 2nd iteration than at the 1st iteration but was greatest at the 3rd iteration. Unlike the CALF metrics, two variables for task emotions showed a more complex pattern of change. The participants' anxiety was greatest at the 1st iteration, decreased at the 2nd iteration, but slightly increased at the 3rd iteration. Their enjoyment was lowest at the 1st iteration, increased at the 2nd iteration, but

slightly decreased at the 3rd iteration. Whether these changes are significant was statistically investigated for the second research question, and the results will be described later in this section.

After checking the descriptive statistics of the variables, we then check if the random assignment was successfully carried out before further data analyses. To this end, we ran three ANOVA tests for the placement test, age, and gender variables to see if the participants assigned to three groups were statistically similar to each other.

As shown in Table 3, we found that the participants assigned to each treatment group had similar means and standard deviations for all three variables, such as the placement test, age, and gender variables. In addition, the results of the ANOVA tests indicated that the seemingly minor differences in the placement test, age, and gender variables across groups were not statistically significant.

Table 2
Descriptive statistics for variables at each iteration for all participants

Variable		**1st iteration**			**2nd iteration**			**3rd iteration**		
		Mean	**Min**	**Max**	**Mean**	**Min**	**Max**	**Mean**	**Min**	**Max**
Syntactic complexity	MLT	16.18 (2.15)	10.33	26.45	16.45 (2.22)	10.02	23.61	16.82 (2.23)	10.82	27.73
	DCT	0.64 (0.15)	0.21	1.00	0.66 (0.13)	0.26	0.98	0.66 (0.13)	0.24	0.98
	CNT	1.63 (0.24)	0.68	2.56	1.65 (0.24)	1.08	2.71	1.67 (0.28)	1.02	3.25
Accuracy	EFCs	0.57 (0.11)	0.14	0.99	0.59 (0.11)	0.33	0.97	0.61 (0.12)	0.23	0.97
	CVFs	0.57 (0.11)	0.23	0.99	0.59 (0.10)	0.32	0.98	0.62 (0.12)	0.38	0.99
Lexical complexity	MTLD	69.38 (11.24)	32.12	109.60	70.79 (10.93)	41.17	98.70	72.41 (10.48)	43.61	99.76
	WRDFRQmc	1.66 (0.16)	1.11	1.98	1.67 (0.15)	1.09	1.99	1.69 (0.13)	1.09	1.98
Fluency	WM	11.05 (1.26)	7.54	16.63	10.95 (0.82)	9.23	14.90	11.02 (1.00)	8.93	18.29
Emotion	Anxiety (6 items)	3.35 (1.21)	1.00	5.67	3.01 (0.93)	1.17	5.50	3.23 (0.92)	1.50	5.67
	Enjoyment (5 items)	3.50 (1.25)	1.20	5.80	3.86 (0.93)	1.40	5.80	3.65 (0.94)	1.40	5.80

Values in parentheses are standard deviations. MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; and W/M: words per minute

Table 3
Homogeneity checks between treatment groups

	Long-spaced	Short-spaced	Massed	ANOVA	p-value
Placement test	7.66 (0.38)	7.70 (0.36)	7.69 (0.39)	F = 0.26	0.774
Age	19.96 (1.39)	20.08 (1.40)	20.00 (1.42)	F = 0.17	0.840
Gender	0.61 (0.49)	0.53 (0.50)	0.56 (0.50)	F = 0.58	0.558

Values are means and those in parentheses are standard deviations. MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment

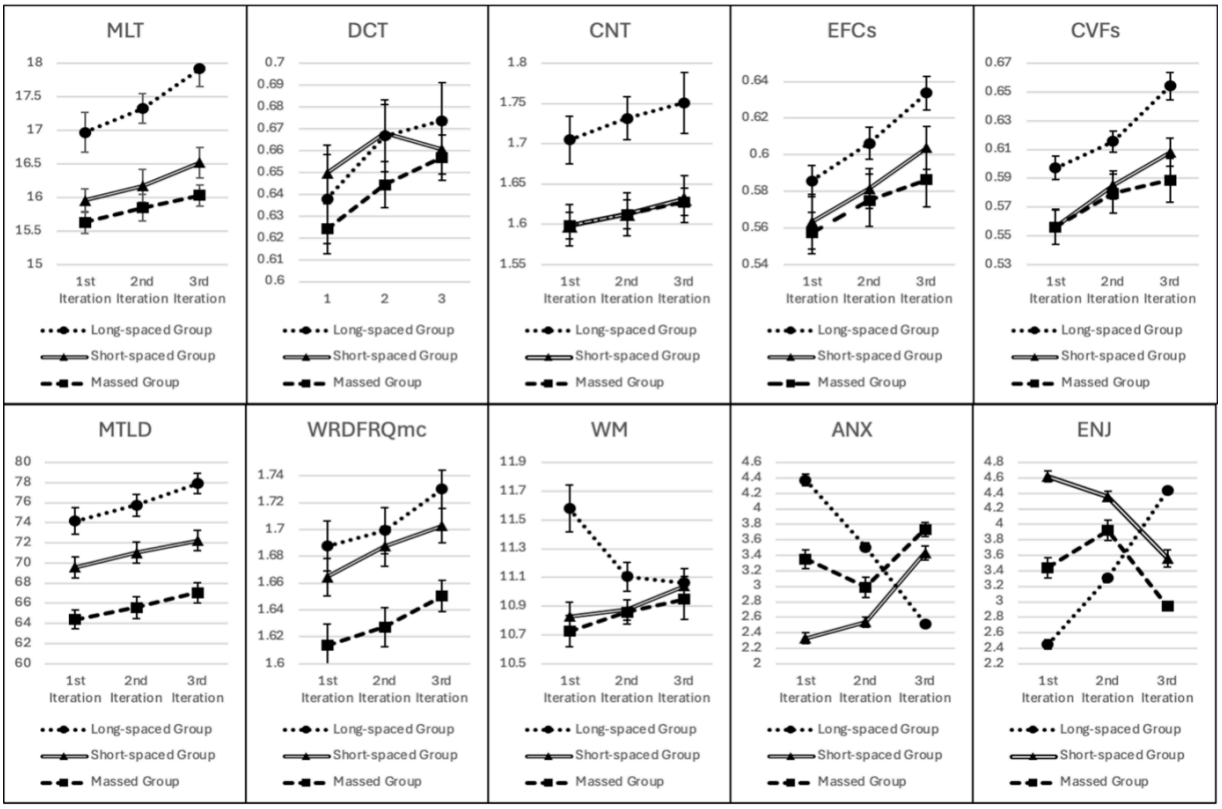


Fig. 1 Changes in task performance and emotions across three iterations for three task repetition conditions

When undertaking different task repetition types, however, participants showed different patterns of increasing and decreasing their task performance and emotions. To visualize these differences, we included 10-line graphs for each measurement in Fig. 1 (Readers refer to Table 6 through 11 in the following section for detailed descriptive statistics of the variables for each task repetition condition at the 1st, 2nd, and 3rd iterations). For example, we could identify that (1) participants showed overall increments for their syntactic complexity (MLT, DCT, CNT), accuracy (EFCs, CVFs), and lexical complexity (MTLD, WRDFRQmc), (2) participants in the Long-spaced group had seemingly higher scores for MLT, CNT, EFCs,

CVFs, and MTLD than others in the Short-spaced and Massed groups after the experiment (i.e., at the 3rd iteration), (3) participants in the Short-spaced group had higher scores for MTLD than those in the Massed group, (4) participants' fluency (WM) became similar with each other after the experiment (i.e., at the 3rd iteration), (5) participants in the Long-spaced group had lower anxiety and higher enjoyment than those in the Short-spaced and Massed group after the experiment (i.e., at the 3rd iteration), and (6) participants in the Short-spaced group had lower anxiety and higher enjoyment than those in the Massed group after the experiment (i.e., at the 3rd iteration). In the following sections, we report results from further data analysis to answer the research questions.

RQ 1: Impact of task repetition schedules on L2 writing task performance and task emotions

After confirming that the random assignment was successfully carried out, we then conducted regression analyses to investigate the impact of providing different task repetition schedules on learners' L2 writing task performance and emotion.

First, regarding task performance in terms of syntactic complexity, participants undergoing long-spaced task repetitions outperformed their peers in the short-spaced repetition group after one repetition (i.e., at the 2nd iteration) for MLT and CNT measurements (see Table 4). Differences in performance between participants in short-spaced and massed repetition groups were minimal and statistically insignificant. However, for DCT measurements of syntactic complexity and accuracy (EFC and CVFs), participants showed consistent performance across all task repetition schedules. Regarding lexical complexity, particularly MTLD, participants in the long-spaced repetition group exhibited the highest performance, followed by those in the short-spaced and massed repetition groups. Performance differences among these groups were statistically significant. Unlike MTLD, participants in the long- and short-spaced repetition groups showed similar levels of performance, which were significantly higher than those in the massed repetition group. Lastly, fluency showed consistent performance across all task repetition schedules.

Regarding task emotion, participants undergoing long-spaced task repetitions reported higher anxiety and lower enjoyment compared to their counterparts in the short-spaced repetition group after one repetition (i.e., at the 2nd iteration). Similarly, those in the massed repetition group exhibited higher anxiety and lower enjoyment than their peers in the short-spaced repetition group. Consequently, participants in the short-spaced task repetition group experienced the lowest anxiety levels and highest enjoyment across the three groups.

Second, examining the 3rd iteration to assess the impact of different task repetition schedules after two repetitions revealed a pattern similar to our findings after one repetition. As indicated in Table 5, participants in the long-spaced task repetition schedule outperformed their peers in short-spaced and massed repetition groups after two repetitions (i.e., at the 3rd iteration) for MLT and CNT measurements. Differences in performance between participants in short-spaced and massed repetition groups remained minor and statistically insignificant. However, for DCT measurements of syntactic complexity and accuracy (EFC and CVFs), participants showed consistent performance across all task repetition schedules. Regarding lexical complexity, particularly MTLD, participants in the long-spaced repetition group showed the highest performance, followed by those in the short-spaced and massed repetition. Performance differences among these groups were statistically significant. Unlike MTLD, participants in the long- and short-spaced repetition groups exhibited similar levels of performance, which were significantly higher than those in the massed repetition group. Lastly, fluency demonstrated consistent performance across all task repetition schedules.

Table 4

Impact of task repetition schedules on writing after one repetition (at the 2nd Iteration)

REPEATED once	**Syntactic complexity**			**Accuracy**		**Lexical complexity**		**Fluency**	**Emotion**	
	MLT	**DCT**	**CNT**	**EFCs**	**CVFs**	**MTLD**	**WRDFRQmc**	**WM**	**Anxiety**	**Enjoyment**
Long-spaced	0.21**	-0.00	0.21**	0.07	0.09	0.18**	0.03	0.13	0.82***	-1.06***
	(0.32)	(0.02)	(0.03)	(0.02)	(0.01)	(1.53)	(0.02)	(0.13)	(0.17)	(0.15)
Short-spaced	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)
Massed	-0.06	-0.06	-0.00	-0.02	-0.03	-0.20**	-0.19**	-0.01	0.39***	-0.51***
	(0.32)	(0.02)	(0.03)	(0.02)	(0.01)	(1.53)	(0.02)	(0.12)	(0.13)	(0.12)
Post-hoc comparison	L > S	L≈S	L > S	L≈S	L≈S	L > S	L≈S	L≈S	L > S	L < S
	S≈M	S≈M	S≈M	S≈M	S≈M	S > M	S > M	S≈M	M > S	S > M
	L > M	L≈M	L > M	L≈M	L > M	L > M	L > M	L≈M	L > M	L < M

Values are standardized coefficients, and those in parentheses are standard errors. All estimates were computed after controlling for task performance at the 1st iteration as well as other control variables. In this regression model, the short-spaced group serves as the reference group, allowing comparisons between the long-spaced and short-spaced groups, as well as the short-spaced and massed groups. Comparisons between the long-spaced and massed groups were conducted by changing the reference group within the regression framework. The results of these comparisons are summarized in the 'Post-hoc Comparison' row. For example, 'L > S' indicates that the long-spaced group had significantly higher values than the short-spaced group, while 'S≈M' indicates no statistically significant difference between the short-spaced and massed groups. MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; and W/M: words per minute. ** $p < 0.01$

Table 5

Impact of task repetition schedules on writing after two repetitions (at the 3rd Iteration)

Repeated twice	Syntactic complexity			Accuracy		Lexical complexity		Fluency	Emotion	
	MLT	DCT	CNT	EFCs	CVFs	MTLD	WRDFRQmc	WM	Anxiety	Enjoyment
Long-spaced	0.31***	0.05	0.22**	0.10	0.12	0.22***	0.09	-0.00	-0.65***	0.57***
	(0.32)	(0.02)	(0.04)	(0.02)	(0.02)	(1.44)	(0.02)	(0.16)	(0.16)	(0.15)
Short-spaced	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)
Massed	-0.10	-0.00	-0.00	-0.07	-0.08	-0.20**	-0.17*	-0.04	0.06	-0.24***
	(0.31)	(0.02)	(0.04)	(0.02)	(0.02)	(1.44)	(0.02)	(0.15)	(0.12)	(0.12)
Post-hoc comparison	L > S S≈M L > M	L≈S S≈M L≈M	L > S S≈M L > M	L≈S S≈M L > M	L≈S S≈M L > M	L > S S > M L > M	L≈S S > M L > M	L≈S S≈M L≈M	L < S S≈M L < M	L > S S > M L > M

Values are standardized coefficients, and those in parentheses are standard errors. All estimates were computed after controlling for task performance at the 1st iteration as well as other control variables. In this regression model, the short-spaced group serves as the reference group, allowing comparisons between the long-spaced and short-spaced groups, as well as the short-spaced and massed groups. Comparisons between the long-spaced and massed groups were conducted by changing the reference group within the regression framework. The results of these comparisons are summarized in the 'Post-hoc Comparison' row. For example, 'L > S' indicates that the long-spaced group had significantly higher values than the short-spaced group, while 'S≈M' indicates no statistically significant difference between the short-spaced and massed groups. MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; and W/M: words per minute. *** $p < .001$, ** $p < 0.01$, * $p < .05$

When it comes to emotions, participants in the long-spaced task repetition group reported lower anxiety and higher enjoyment compared to those in the short-spaced repetition group after two repetitions (i.e., at the 3rd iteration). Conversely, participants in the short-spaced task repetition group, who initially showed the lowest anxiety and highest enjoyment after one repetition, reported similar levels of anxiety and lower enjoyment compared to their peers in the long-spaced repetition group. Consequently, participants in the massed repetition group exhibited the highest levels of anxiety and lowest enjoyment.

RQ 2a: Changes of task performance and task emotions after one repetition

While the first research question focused on measuring the impact of different task repetition schedules on L2 writing task performance and emotion at specific periods, such as after one and two repetitions, the second research question explored the impact on changes in task performance and emotion across different time points.

Table 6

Long-spaced group's changes of task performance after one repetition between 1st and 2nd iterations

Long-spaced (1st iteration Vs. 2nd iteration)		**1st Iteration**		**2nd Iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	16.97	2.79	17.33	2.13	1.10	0.275
	DCT	0.64	0.19	0.67	0.16	1.31	0.193
	CNT	1.70	0.28	1.73	0.25	0.78	0.436
Accuracy	EFCs	0.59	0.08	0.61	0.08	2.09	0.039
	CVFs	0.60	0.08	0.62	0.07	1.89	0.061
Lexical complexity	MTLD	74.16	12.53	75.76	10.26	0.95	0.344
	WRDFRQmc	1.69	0.18	1.70	0.16	0.43	0.665
Fluency	WM	11.58	1.54	11.11	0.96	-2.47	0.015
Emotion	Anxiety	4.37	0.75	3.51	0.49	-11.17*	0.000
	Enjoyment	2.45	0.56	3.31	0.35	12.5*	0.000

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; and W/M: words per minute. Although our analysis consists of independent comparisons across 10 different measurements, we applied the Bonferroni correction to ensure the robustness of my findings and maintain the overall significance level (e.g., $p < 0.005$ for statistical significance instead of 0.05; i.e., 0.05/10 = 0.005). Degrees of freedom (df) for all tests = 269. * $p < .005$

First, for participants in the long-spaced task repetition group (see Table 6), we found that they showed higher scores only for the EFCs metric in the accuracy aspect. For the remaining metrics, participants showed slightly higher (not significant) performance except for fluency. For fluency, participants had lower (not significant) scores for WM after one repetition at the 2nd iteration. Concerning emotions, participants in the long-spaced task experienced significantly decreased anxiety and increased enjoyment.

For participants in the short-spaced task repetition group (see Table 7), we found that they showed higher (not significant) scores for the CVFs metric in the accuracy aspect. For the remaining metrics, participants did not show any statistically significant changes. For their emotions, however, they became significantly more anxious and experienced less enjoyment after their first task repetition.

For participants in the massed task repetition group (see Table 8), we found that participants did not show any statistically significant changes across all metrics, including task performance and emotions.

In short, our investigation of the changes in task performance after one repetition between the 1st and 2nd iterations indicated that (1) participants in the long- and short-spaced task repetition conditions experienced some improvements in the accuracy of their L2 writing, (2) participants in the long-spaced task repetition condition had less fluent writing performance after one repetition, and (3) participants in the long-spaced task repetition condition

experienced decreased anxiety and increased enjoyment while those in the short-spaced task repetition condition experienced the opposite emotional change (i.e., increased anxiety and decreased enjoyment).

Table 7

Short-spaced group's changes of task performance after one repetition between 1st and 2nd iterations

Short-spaced (1st iteration vs. 2nd iteration)		**1st iteration**		**2nd iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	15.95	1.64	16.17	2.36	0.73	0.465
	DCT	0.65	0.12	0.67	0.12	1.34	0.184
	CNT	1.60	0.25	1.61	0.25	0.36	0.722
Accuracy	EFCs	0.56	0.14	0.58	0.10	1.19	0.236
	CVFs	0.56	0.12	0.59	0.09	1.96	0.053
Lexical complexity	MTLD	69.57	9.68	71.03	9.98	1.20	0.232
	WRDFRQmc	1.66	0.13	1.69	0.14	1.26	0.211
Fluency	WM	10.83	0.94	10.87	0.66	0.41	0.685
Emotion	Anxiety	2.33	0.67	2.54	0.60	4.04*	0.000
	Enjoyment	4.62	0.67	4.36	0.61	-4.87*	0.000

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; and W/M: words per minute. Although our analysis consists of independent comparisons across 10 different measurements, we applied the Bonferroni correction to ensure the robustness of my findings and maintain the overall significance level (e.g., $p < 0.005$ for statistical significance instead of 0.05; i.e., 0.05/10 = 0.005). Degrees of freedom (df) for all tests = 269. * $p < .005$

Table 8

Massed group's changes of task performance after one repetition between 1st and 2nd iterations

Massed (1st iteration vs. 2nd iteration)		**1st Iteration**		**2nd Iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	15.63	1.58	15.84	1.88	0.90	0.372
	DCT	0.62	0.11	0.64	0.10	1.47	0.146
	CNT	1.60	0.15	1.61	0.17	0.61	0.543
Accuracy	EFCs	0.56	0.11	0.58	0.13	1.24	0.216
	CVFs	0.56	0.11	0.58	0.13	1.64	0.105
Lexical complexity	MTLD	64.4	9.12	65.59	10.18	0.90	0.371
	WRDFRQmc	1.61	0.15	1.63	0.14	0.63	0.532
Fluency	WM	10.73	1.05	10.86	0.82	0.92	0.362
Emotion	Anxiety	3.35	1.15	2.99	1.24	-1.53	0.130
	Enjoyment	3.44	1.24	3.92	1.25	1.88	0.064

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment. Although our analysis consists of independent comparisons across 10 different measurements, we applied the Bonferroni correction to ensure the robustness of my findings and maintain the overall significance level (e.g., $p < 0.005$ for statistical significance instead of 0.05; i.e., 0.05/10 = 0.005). Degrees of freedom (df) for all tests = 269

RQ 2b: Changes in task performance and task emotions after two repetitions

After investigating changes in task performance after one repetition (i.e., between the 1st and 2nd), we then focused on how our findings would differ in situations involving two repetitions (i.e., between the 1st and 3rd iterations).

Regarding the long-spaced task repetition condition, as shown in Table 9, participants achieved higher performance in the MLT (not significant), DCT (not significant), and CNT (not significant) components of syntactic complexity. They also performed significantly better in the EFCs and CVFs components of accuracy and the MTLD (not significant) and WRDFRQmc (not significant) components of lexical complexity. However, for the WM component of fluency, the participants showed a decrease (not significant) after two repetitions. Regarding their emotions, we found that participants in the long-spaced group became significantly less anxious and found the task more enjoyable.

However, participants in the short-spaced task repetition condition demonstrated slightly different patterns of task performance from their peers in the long-spaced task repetition condition. Above all, although the difference was not significant, participants showed slightly higher performance in fluency (not significant) after two repetitions (see Table 16). Other than that, we found that participants achieved higher performance in the MLT (not significant), DCT (not significant), and CNT (not significant) components of syntactic complexity; the EFCs and CVFs components of accuracy; and the MTLD and WRDFRQmc components of lexical complexity. See Table 10 for details. Concerning their emotions, participants in the short-spaced task repetition condition became more anxious and found the task less enjoyable after the two repetitions.

Table 9

Long-spaced group's changes of task performance after two repetitions between 1st and 3rd iterations

Long-spaced (1st iteration vs. 3rd iteration)		**1st Iteration**		**3rd Iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	16.97	2.79	17.92	2.49	2.39	0.019
	DCT	0.64	0.19	0.67	0.17	1.40	0.166
	CNT	1.70	0.28	1.75	0.36	0.93	0.356
Accuracy	EFCs	0.59	0.08	0.63	0.09	4.38*	0.000
	CVFs	0.60	0.08	0.65	0.09	4.91*	0.000
Lexical complexity	MTLD	74.16	12.53	77.91	9.65	2.28	0.025
	WRDFRQmc	1.69	0.18	1.73	0.14	1.94	0.066
Fluency	WM	11.58	1.54	11.06	0.95	-2.64	0.010
Emotion	Anxiety	4.37	0.75	2.51	0.41	-27.40*	0.000
	Enjoyment	2.45	0.56	4.44	0.38	37.93*	0.000

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment. Although our analysis consists of independent comparisons across 10 different measurements, we applied the Bonferroni correction to ensure the robustness of my findings and maintain the overall significance level (e.g., $p < 0.005$ for statistical significance instead of 0.05; i.e., 0.05/10 = 0.005). Degrees of freedom (df) for all tests = 269. * $p < .005$

Table 10

Short-spaced group's changes of task performance after two repetitions between 1st and 3rd iterations

Short-spaced (1st iteration vs. 3rd iteration)		**1st iteration**		**3rd iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	15.95	1.64	16.52	2.17	1.84	0.068
	DCT	0.65	0.12	0.66	0.11	0.81	0.423
	CNT	1.60	0.25	1.63	0.28	0.74	0.463
Accuracy	EFCs	0.56	0.14	0.60	0.11	2.25*	0.027
	CVFs	0.56	0.12	0.61	0.09	3.62***	0.000
Lexical complexity	MTLD	69.57	9.68	72.24	9.56	2.35*	0.021
	WRDFRQmc	1.66	0.13	1.70	0.12	2.32*	0.022
Fluency	WM	10.83	0.94	11.04	0.61	1.86	0.066
Emotion	Anxiety	2.33	0.67	3.43	0.89	17.38***	0.000
	Enjoyment	4.62	0.67	3.56	1.04	-14.20***	0.000

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment. Degrees of freedom (df) for all tests = 269

*** $p < .001$, * $p < .05$

For participants in the massed repetition task condition, we found a similar pattern of task performance with the short-spaced repetition task condition. To be brief, participants in this condition achieved higher performance after two repetitions across all metrics. As shown in Table 11, we found that participants achieved higher performance in the MLT (not significant),

DCT, and CNT (not significant) components of syntactic complexity; the EFCs (not significant) and CVFs components of accuracy; the MTLD and WRDFRQmc components of lexical complexity; and the WM (not significant) component of fluency. Concerning emotions, participants in the massed repetition task condition became more anxious and found the task less enjoyable.

In short, our investigation of the changes in task performance after two repetitions between the 1st and 3rd iterations indicated that (1) all participants achieved higher performance in syntactic complexity, accuracy, and lexical complexity, (2) participants in the long-spaced task repetition condition showed significantly lower performance in fluency after two repetitions, and (3) participants in the long-spaced task repetition condition became less anxious and found the task more enjoyable while those in the other conditions became more anxious and found the tasks less enjoyable.

Table 11

Massed group's changes of task performance after two repetition between 1st and 3rd iterations

Massed (1st iteration vs. 3rd iteration)		**1st Iteration**		**3rd Iteration**		**Improvement**	
		M	**SD**	**M**	**SD**	**t-statistics**	**p-value**
Syntactic complexity	MLT	15.63	1.58	16.03	1.50	1.82	0.072
	DCT	0.62	0.11	0.66	0.10	2.38*	0.019
	CNT	1.60	0.15	1.63	0.16	1.66	0.101
Accuracy	EFCs	0.56	0.11	0.59	0.14	1.82	0.072
	CVFs	0.56	0.11	0.59	0.14	2.48*	0.015
Lexical complexity	MTLD	64.4	9.12	67.06	9.37	2.08*	0.041
	WRDFRQmc	1.61	0.15	1.65	0.11	1.98*	0.051
Fluency	WM	10.73	1.05	10.95	1.32	1.35	0.182
Emotion	Anxiety	3.35	1.15	3.74	0.88	2.31*	0.023
	Enjoyment	3.44	1.24	2.94	0.58	-2.97***	0.004

MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment. Degrees of freedom (df) for all tests = 269

* $p < .05$

RQ 2c. Effect of task repetition schedules on task performance and task emotions across different time points

In this section, we report the results of the time-series cross-sectional analysis, which included interaction terms between task repetition schedules and iterations to explore how the effect of task repetition schedules on task performance varies over different periods. We had three task schedules: long-spaced (group 1), short-spaced (group 2), and massed (group 3) repetition schedules. There were also three time periods: the 1st (time 1), 2nd (time 2), and 3rd (time 3)

iterations. The analysis results involved coefficients for group 1##c.time, group 2##c.time, and group 3##c.time, representing how the effect of each task repetition schedule on task performance changes across time points 1, 2, and 3.

As shown in Table 12, the results revealed only one statistically significant coefficient, when it comes to task performance. This indicates that while the effects of different task repetition schedules on task performance changed over time, they were not significantly different between the task repetition schedules for the syntactic complexity, accuracy, and lexical complexity aspects, except for the fluency aspect. More specifically, this indicated that participants in the long-spaced repetition task condition experienced a more negative change in their fluency over time compared to their peers in the other groups, with this negative change being 0.35 standard deviations.

When it comes to task emotions, particularly for the anxiety aspect, participants in the long-spaced task repetition schedule showed a more positive change over time compared to their counterparts in the short-spaced repetition group, with this change measuring 1.43 standard deviations. Similarly, those in the short-spaced repetition task experienced a more positive change in anxiety compared to those in the massed group, with a change of 0.35 standard deviations. Combining these findings, participants in the long-spaced group experienced a total positive change in anxiety of 1.78 standard deviations compared to those in the massed group.

Turning to the enjoyment aspect of task emotions, a similar but opposite pattern emerged. Participants in the long-spaced repetition task reported a more positive change in enjoyment over time compared to those in the short-spaced task repetition group, with this change being 1.44 standard deviations. Likewise, participants in the short-spaced repetition task showed a more positive change in enjoyment compared to the massed group, with a change of 0.27 standard deviations. When combined, participants in the long-spaced group experienced an overall positive change in enjoyment of 1.71 standard deviations compared to those in the massed group.

Table 12

Effect of task repetition schedules on task performance across different time points

Group × Time	Syntactic Complexity			Accuracy		Lexical Complexity		Fluency	Emotion	
	MLT	DCT	CNT	EFCs	CVFs	MTLD	WRDFRQmc	WM	Anxiety	Enjoyment
Long-spaced	0.09	0.09	0.03	0.03	0.02	0.05	0.01	-0.35**	-1.43***	1.44***
	(0.11)	(0.11)	(0.13)	(0.09)	(0.08)	(0.09)	(0.09)	(0.11)	(0.04)	(0.04)
Short-spaced	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)	(ref)
Massed	-0.04	0.08	-0.01	-0.05	-0.09	-0.00	-0.00	0.01	-0.35***	0.27***
	(0.09)	(0.07)	(0.09)	(0.11)	(0.09)	(0.08)	(0.08)	(0.10)	(0.09)	(0.09)
Post-hoc comparison	L≈S	L≈S	L≈S	L≈S	L≈S	L≈S	L≈S	L < S	L < S	L > S
	S≈M	S≈M	S≈M	S≈M	S≈M	S≈M	S≈M	S≈M	S > M	S < M
	L≈M	L≈M	L≈M	L≈M	L≈M	L≈M	L≈M	L < M	L < M	L > M

Values are standardized coefficients, and those in parentheses are standard errors. All estimates were computed after controlling for task performance at the 1st iteration as well as other control variables. In this regression model, the short-spaced group serves as the reference group, allowing comparisons between the long-spaced and short-spaced groups, as well as the short-spaced and massed groups. Comparisons between the long-spaced and massed groups were conducted by changing the reference group within the regression framework. The results of these comparisons are summarized in the 'Post-hoc Comparison' row. For example, 'L > S' indicates that the long-spaced group had significantly higher values than the short-spaced group, while 'S≈M' indicates no statistically significant difference between the short-spaced and massed groups. MLT: the mean length of T-unit; DC/T: the ratio of dependent clauses to T-units; CN/T: the ratio of complex nominals per T-unit; EFCs: error-free clauses; CVFs: correct verb forms; MTLD: the measure of textual lexical diversity; WRDFRQmc: the log frequency of content words; W/M: words per minute; ANX: anxiety; and ENJ: enjoyment. ** $p < 0.01$

Discussion

Effects of different schedules of task repetition on l2 writing task performance and task emotions

Task performance

Participants in the long-spaced repetition group consistently outperformed those in the short-spaced and massed repetition groups in terms of syntactic complexity (measured by MLT and CNT). This superiority can be attributed to the practice effect encompassing the spacing and lag effects (Rogers, 2017). According to the practice effect, spaced practice with a long lag allows for more effective encoding and consolidation of information into long-term memory (Cepeda et al., 2006). Recalling what they have previously done with a relatively long interval seems to enable participants in the long-spaced repetition group more effectively to integrate syntactic rules, resulting in enhanced syntactic complexity in their writing task performance. Conversely, the minimal differences observed between the short-spaced and massed repetition groups revealed that shorter intervals did not provide significant additional benefits over massed repetition for syntactic complexity. Massed repetition, characterized by intense, closely spaced practice sessions, appears to lead to shallow encoding and limited retention. Participants in these groups may engage in rote memorization rather than deep learning, which could explain their lower performance in syntactic complexity compared to the long-spaced repetition group.

Across all task repetition schedules (long-spaced, short-spaced, and massed), participants consistently showed similar performance in both accuracy metrics (EFCs and CVFs). This consistency suggests that the participants in the current study were proficient enough in producing accurate structures regardless of the interval between practice sessions. This stability in L2 writing accuracy across different task repetition schedules suggests that once participants achieved a certain level of syntactic proficiency, their ability to maintain accuracy was resilient to variations in the spacing of practice sessions. Therefore, while spaced repetition schedules may enhance syntactic complexity through deeper encoding and consolidation, they appear to have less impact on the core ability to produce accurate structures (Smith & Scarf, 2017).

Regarding lexical complexity, participants in the long-spaced repetition group achieved the highest performance in MTLD, indicating greater lexical diversity in their writing. This finding aligns with research on spaced learning, which suggests that longer intervals between practice sessions facilitate deeper semantic processing and more effective retrieval of lexical items (Gordon, 2020). In the context of lexical learning, this means that participants in the long-spaced repetition group had more time to engage in semantic elaboration and association, which are crucial processes for developing a rich and varied lexicon. Finally, performance in L2 written fluency remained consistent across all task repetition schedules, indicating that while spaced repetition may enhance syntactic and lexical complexity, it does not necessarily impact the overall speed or fluency of language production. Fluency in writing tasks depends more on automaticity and real-time production practice rather than the intervals between spaced learning sessions (Gánem-Gutiérrez & Gilmore, 2018).

Task emotions (anxiety and enjoyment)

When examining task-related emotions, we observed somewhat inconsistent patterns between the findings after one repetition and those after two repetitions. For example, following the first repetition of the writing task, participants in the short-spaced repetition group reported the lowest anxiety levels and highest enjoyment. However, after the second repetition, participants in the long-spaced repetition group exhibited the lowest anxiety levels and highest enjoyment.

The finding after the one repetition indicated that shorter intervals between repetitions had a positive impact on learners' emotional experience during the writing tasks. In fact, shorter intervals between practice sessions allow learners to maintain a sense of familiarity with the tasks. This familiarity stems from more frequent exposure to the material, which helps learners feel more confident in their ability to handle the challenges presented by writing tasks (Abdi

Tabari et al., 2024). When learners feel familiar with the content and tasks, they are less likely to experience anxiety related to uncertainty or unfamiliarity. This reduced anxiety level can create a more comfortable and supportive learning environment (Güvendir & Uzun, 2023).

However, the other side of the findings showed that participants in the long-spaced repetition group were able to overcome cognitive load and uncertainty coming from longer intervals between practice sessions (de Jong, 2009) after the two repetitions of the writing task. It has been argued that when learners practice at long intervals, they may struggle to recall and apply previously learned material because the information is not as fresh in their minds. This difficulty can lead to heightened anxiety as learners fear they might forget or misapply what they have learned. Additionally, uncertainty about their performance in less frequently practiced tasks can contribute to lower enjoyment and satisfaction with the learning process (Abuhamdeh et al., 2014). Nevertheless, we could identify that when practice sessions are repeated adequately, they may feel sure about their abilities and progress, which can positively affect their motivation and overall enjoyment. Finally, while massed repetition might initially seem beneficial due to the sheer volume of practice, it can lead to fatigue and decreased enjoyment over time (Brown et al., 2014). Learners may feel overwhelmed by the amount of material they need to cover in each iteration, without adequate time for consolidation and mastery. This can result in frustration and a sense of being overwhelmed, further reducing enjoyment in the learning process (YeckehZaare et al., 2019).

Changes of task performance and task emotions after one repetition

For participants in the long-spaced task repetition group, significant findings emerged in both task performance and emotions. Specifically, performance improvements were noted in effective writing characteristics (EFCs), indicating enhanced writing accuracy. However, fluency, as measured by writing mechanics (WM), significantly declined after one repetition. This may be attributed to a trade-off effect, where learners prioritize accuracy over fluency when the demands of the task exceed their cognitive resources (Skehan, 2009). In this scenario, the longer intervals provided learners with additional time for cognitive processing and reflection, which may have led them to focus more on producing accurate language at the expense of fluidity. This suggests that the spacing of repetitions plays a critical role in shaping learners' strategic focus during L2 writing task performance, enabling them to concentrate on refining their L2 written production.

In terms of task emotions, participants in the long-spaced group experienced a marked decrease in anxiety and a notable increase in enjoyment. The extended interval between repetitions likely allowed for greater cognitive processing and consolidation of information, thereby reducing anxiety and enhancing overall enjoyment (Tahmouresi & Papi, 2021). The

additional time may facilitate a more relaxed learning environment, where learners feel less pressured and more empowered to express themselves in writing. This emotional relief is critical, as lower anxiety levels can lead to improved cognitive resources for L2 writing tasks, ultimately enhancing L2 writing task performance. Higher enjoyment levels can further motivate learners to engage more fully with the writing process, resulting in improved writing outcomes. Conversely, the short-spaced task repetition group showed improvements in correct verb forms (CVFs), which can be attributed to the repetition itself. However, this group also reported increased anxiety and decreased enjoyment. This phenomenon can be explained by cognitive load theory, which posits that frequent repetitions with insufficient breaks can overwhelm working memory, leading to negative emotional outcomes (Atiomo, 2020). In the context of L2 writing, the short intervals likely provided inadequate time for cognitive processing, resulting in heightened stress and decreased emotional satisfaction. Increased anxiety in this group may hinder cognitive resources available for effective writing, negatively impacting overall writing performance. When learners are overwhelmed, their ability to focus on composing coherent and grammatically accurate texts may be compromised, leading to less favorable L2 writing outcomes.

For participants in the massed task repetition group, no statistically significant changes were observed in either task performance or task emotions. This lack of change indicates that massed repetition did not produce improvements or declines in performance metrics or emotional states. This result can be linked to the concept of massed practice, which often leads to immediate performance gains but fails to foster long-term retention or emotional benefits due to the absence of cognitive consolidation time (Smolen et al., 2016). In L2 writing contexts, without opportunities for reflection and processing, massed practice may inhibit the development of deeper writing skills and emotional engagement with the task, preventing learners from realizing their full potential in written expression.

The differences observed across the three groups can be understood through the lens of cognitive load and emotional regulation in relation to L2 writing task performance. In the long-spaced group, the combination of increased processing time and reduced anxiety facilitated greater accuracy and enjoyment in writing tasks, likely leading to improved writing quality. In contrast, the short-spaced and massed groups experienced increased anxiety and stagnation in performance, highlighting the critical need for task repetition schedules that effectively balance cognitive demands with emotional well-being. Understanding how these schedules influence both performance and emotional states is essential for optimizing L2 writing outcomes, enabling learners to achieve their full potential in L2 written communication.

Changes of task performance and task emotions after two repetitions

For participants in the long-spaced task repetition condition, significant improvements were observed in several performance metrics and emotional responses after two repetitions. Specifically, long-spaced repetition led to enhanced syntactic complexity and accuracy, reflecting a deeper internalization of language structures and vocabulary. This enhancement is likely due to the longer intervals between repetitions, which provided participants ample time to consolidate their learning and apply it effectively in subsequent tasks (Nitta & Baba, 2014). However, fluency showed a significant decrease. This decline may again be attributed to the trade-off phenomenon (Skehan, 2009), where learners likely prioritized accuracy and complexity over fluency. These findings suggest that the effects of task repetition schedules on L2 writing task performance are multifaceted, with different linguistic aspects (CALF) developing variably as learners perform tasks multiple times (Hiver, Al-Hoorie, & Murakami, 2024).

In terms of task emotions, participants in this group demonstrated a lower level of anxiety and a higher level of enjoyment. This positive emotional shift aligns with findings that longer intervals between practice sessions can contribute to reduced anxiety and increased satisfaction. The extended time allows participants to manage stress better and approach the L2 writing task with a more relaxed mindset, which in turn fosters their overall writing experience (Shafiee Rad & Jafarpour, 2022). This emotional benefit is crucial, as lower anxiety levels can facilitate better cognitive engagement with the writing task, ultimately enhancing performance and motivation.

Participants in the short-spaced task repetition group exhibited a different pattern after two repetitions. Although improvements in fluency were observed, they were not statistically significant. This lack of significant fluency gains indicates that the short-spaced practice may not provide adequate cognitive relief, which is essential for optimal writing task performance in L2 contexts. While performance in areas such as syntactic complexity, accuracy, and lexical complexity showed some improvements, these were not as pronounced as those seen in the long-spaced group. This suggests that short-spaced repetitions may yield some benefits, but they fall short of facilitating deeper learning and consolidation compared to longer intervals (Gordon, 2020). Regarding task emotions, participants in the short-spaced task repetition group experienced increased anxiety and decreased enjoyment. This emotional response may stem from heightened stress levels associated with the frequent task repetitions and a lack of time to mentally recuperate. Participants may have felt overwhelmed by the same task, leading to boredom and dissatisfaction (de Jong, 2009). This finding underscores the importance of pacing in task repetition for sustaining emotional engagement and positive learning experiences in L2 writing tasks.

Finally, participants in the massed task repetition group exhibited improvements in task performance similar to those observed in the short-spaced group. However, these improvements came with notable increases in anxiety and decreases in enjoyment. The immediate gains in performance can be attributed to the increased frequency of practice sessions, which often leads to enhanced familiarity with the task. However, this format can create a cognitive and emotional strain due to the lack of sufficient breaks. The resulting higher anxiety and reduced enjoyment reflect the potential stress and fatigue from continuous practice without adequate recovery time (Verhoeven & Newell, 2018). This pattern highlights the challenges associated with massed repetition in L2 writing, where short-term performance improvements may not translate into long-term benefits, especially regarding emotional well-being and overall task satisfaction. The differences observed across the three groups emphasize the need for careful consideration of task repetition schedules in L2 writing instruction. While long-spaced repetitions appear to support both performance and positive emotional states, short-spaced and massed repetitions may inadvertently hinder L2 learner engagement and writing task performance due to increased cognitive load and emotional strain.

Effect of task repetition schedules on task performance and task emotions across different time points

The findings revealed that the impact of repetition schedules on task performance varied primarily concerning fluency. Participants who engaged in long-spaced repetitions exhibited a notable decline in fluency over time compared to those in the short-spaced and massed conditions. This decline highlights a potential trade-off in L2 writing: while longer intervals may foster deeper cognitive processing and lead to improvements in syntactic complexity and accuracy, they may simultaneously hinder the fluidity of written production. The longer intervals between practice sessions require learners to focus intensively on consolidating their understanding of language structures, which can disrupt the automaticity necessary for fluent writing. This phenomenon reflects the cognitive load involved in task execution and suggests that as learners strive for accuracy and complexity, their fluency may suffer in the short term (Skehan, 2009). In terms of L2 writing metrics, while fluency declined, other aspects, such as syntactic complexity, accuracy, and lexical complexity, remained stable across different repetition schedules. This relative stability indicates that although spacing impacts fluency negatively, it does not uniformly affect all linguistic dimensions. The findings suggest that while spaced repetitions may promote certain linguistic gains, they require careful consideration of how these changes interact with the demands of L2 writing tasks (Bui et al., 2019).

Regarding task emotions, the results highlighted distinct patterns. Participants in the long-spaced repetition group experienced a significant anxiety reduction compared to those in the short-spaced and massed conditions. This decrease in anxiety is crucial, as heightened emotional distress can impede cognitive processes essential for L2 writing performance. Longer intervals likely afforded participants the opportunity to recuperate mentally, manage stress effectively, and approach subsequent writing tasks with greater confidence and composure. This positive emotional shift is reflected in the participants' increased enjoyment levels, indicating that the long-spaced approach not only supports writing performance but also enhances learners' overall engagement and satisfaction with the writing process. In contrast, participants in the short-spaced condition showed a moderate reduction in anxiety and a slight increase in enjoyment, but these changes were less pronounced than those in the long-spaced group. This suggests that while short-spaced repetitions may offer some emotional relief, they fall short of providing the substantial cognitive and emotional support necessary for optimal L2 writing outcomes. Participants in the massed schedule reported the least favorable changes in anxiety and enjoyment, reflecting the negative emotional impact of closely packed practice sessions. The findings imply that massed repetitions can lead to cognitive overload, resulting in higher anxiety levels and diminished writing enjoyment, which ultimately detracts from the learning experience.

Overall, the results indicate that long-spaced repetitions can effectively reduce anxiety and enhance enjoyment, thereby creating a more conducive environment for L2 writing development. However, this approach may also lead to temporary declines in fluency, underscoring the importance of pacing in task repetition. In contrast, while shorter and massed repetition schedules might yield immediate improvements in certain performance metrics, they can adversely affect emotional responses and overall enjoyment of the writing task. These findings highlight the need for educators to carefully design task repetition schedules that balance cognitive demands with emotional well-being, ensuring that learners can achieve optimal performance while maintaining a positive and engaging writing experience. By recognizing the interplay between task performance and emotional factors, instructional strategies can be better aligned with the needs of L2 writers, promoting both their cognitive and emotional development.

Implications, limitations, and directions for future research

The current study supports the theoretical framework of the spacing effect in cognitive psychology, which suggests that spaced practice has beneficial effects, enhancing information encoding and consolidation. The observed improvements in syntactic complexity and lexical diversity in the long-spaced repetition group can align with this theory, indicating that

inserting intervals could promote deeper cognitive processing and better retention of linguistic structures. This finding advocates for incorporating the spacing effect into TBLT research, highlighting its role in optimizing L2 writing outcomes. Additionally, the differential impact of repetition schedules on task-induced emotions reveals the interaction between cognitive processes and emotional responses in L2 writing. The long- and short-spaced repetition group reported lower anxiety and higher enjoyment, suggesting that repeating the task with adequate intervals may improve learners' emotional well-being by reducing uncertainty and fostering a sense of mastery. This underscores the importance of considering emotional factors within TBLT, as task-induced emotions significantly affect L2 writing effectiveness and motivation.

From a pedagogical perspective, the benefits of long-spaced repetition for enhancing syntactic complexity and lexical diversity suggest that educators should design curricula with appropriately spaced intervals between practice sessions. This approach can facilitate deeper learning and retention of complex linguistic structures, ultimately enhancing L2 writing proficiency. Educators may incorporate longer breaks between writing tasks to allow for cognitive consolidation and effective integration of new knowledge. Additionally, it is important to balance cognitive demands with emotional well-being in task repetition schedules. While long-spaced repetition benefits cognitive processing, it may increase anxiety due to longer intervals between practice sessions. Conversely, short-spaced repetition reduces anxiety but may not lead to efficient learning. Therefore, teachers should aim to create repetition schedules that optimize both cognitive and emotional outcomes, possibly by combining elements of both long and short spacing to maintain a supportive L2 writing environment while promoting effective learning. Finally, the stable performance in accuracy across all repetition schedules indicates that factors such as initial proficiency and task familiarity can interact with the effects of writing task repetition. Teachers may focus on learners' initial proficiency and task familiarity before adjusting repetition schedules to enhance syntactic complexity and lexical diversity.

However, the study has some limitations. First, the participant pool was relatively homogeneous, which may limit the generalizability of the findings to more diverse L2 writer populations. Future research should include participants from varied linguistic and cultural backgrounds to enhance the applicability of the results across different writing contexts. Second, the study focused on specific linguistic metrics, including syntactic and lexical complexity, accuracy, and fluency (CALF). While these metrics provide valuable insights into linguistic performance, they do not directly measure global text quality or dimensions such as cohesion and coherence. Future research should consider incorporating direct measures of text quality and additional performance metrics to offer a more comprehensive understanding of how task repetition schedules influence L2 writing. Third, the study did not examine the long-

term retention of learned material beyond the study period. Longitudinal research is needed to evaluate the sustained effects of different repetition schedules on L2 writing performance and task-related emotions, providing insights into the long-term benefits and potential limitations of various scheduling approaches. Despite these limitations, the study offers valuable insights into the effects of task repetition schedules on L2 writing performance and task emotions. These findings highlight the importance of adopting balanced approaches that address both cognitive and emotional factors within the TBLT domain.

Appendix A. Questionnaire Items

Task Anxiety (adapted from Jiang, 2021)

1- When completing the writing task, I felt nervous.

2- When completing the writing task, I worried that the ways I expressed and organized my ideas did not conform to the norms of English writing.

3- When completing the writing task, I felt my heart was pounding.

4- When completing the writing task, I worried that I made language mistakes.

5- As soon as I started completing the writing task, I began to worry about not being able to express myself.

6- When completing the writing task, I got so nervous that I wanted the task to be over.

Task Enjoyment (adapted from Jiang, 2021)

1- The task was interesting.

2- The task was enjoyable.

3- I enjoyed doing the task.

4- The task was exciting.

5- For me, the task was a challenge that is enjoyable.

Declarations

Conflict of interest All authors declare that they have no conflicts of interest.

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